

Homework (Carolina Reaper)

- Q1.** A music festival has two types of tickets: 45 VIP and 30 regular. If the ratio of VIP to regular tickets sold is 3 : 5, what is the total cost of 24 VIP tickets and 40 regular tickets?
(A) 1710
(B) 2160
(C) 1980
(D) 2070
(E) 1890
- Q2.** An artist is creating a painting with a ratio of 2 parts blue paint to 3 parts red paint. If she uses 18 ounces of paint in total, how many ounces of blue paint are used?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12
- Q3.** A music producer is mixing two songs with frequencies of 440 Hz and 880 Hz. If the ratio of the two frequencies is 1 : 2, what is the total frequency of the mixed song?
(A) 1320 Hz
(B) 1100 Hz
(C) 1232 Hz
(D) 1540 Hz
(E) 1760 Hz
- Q4.** An art gallery has two types of paintings: 20
(A) 8
(B) 10
(C) 12
(D) 15
(E) 20
- Q5.** A musician is creating a rhythm pattern with a ratio of 3 parts drum to 2 parts guitar. If the drum plays 18 beats, how many beats does the guitar play?
(A) 8
(B) 10
(C) 12
(D) 14
(E) 16
- Q6.** A craftsman is making a piece of jewelry with a ratio of 2 parts silver to 5 parts gold. If he uses 14 ounces of gold, how many ounces of silver are used?
(A) 4
(B) 5.6
(C) 6
(D) 7
(E) 8
- Q7.** An artist is mixing two types of paint with concentrations of 20
(A) $x = 10$
(B) $x = 15$
(C) $x = 20$
(D) $x = 25$
(E) $x = 30$
- Q8.** A music teacher is creating a playlist with a ratio of 4 parts classical to 3 parts jazz. If the classical music plays for 24 minutes, how many minutes does the jazz music play?
(A) 12
(B) 14
(C) 16
(D) 18
(E) 20

Open Ended Questions (FRQ)

- Q9.** A painter is mixing two types of paint with ratios of 1 : 3 and 2 : 5. If she mixes x ounces of the first paint with y ounces of the second paint, what is the ratio of the final mixture if $x = 10$ and $y = 15$? Show your work and explain your reasoning.
- Q10.** A musician is creating a rhythm pattern with a ratio of 2 parts drum to 3 parts bass. If the drum plays x beats and the bass plays y beats, what is the total number of beats if the ratio of x to y is 4 : 6? Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Make sure to read the questions carefully and understand what is being asked.
- Tip 2: Use the given information to set up equations and solve for the unknowns.
- Tip 3: Check your work and explain your reasoning for the free response questions.